3. What would be the new variance if we added 1 to each element in the dataset $\mathcal{D} = \{1, 2, 3, 2\}$ from Question 1? Please use decimal numbers in your answer.

1 point

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4. What would be the new variance if we multiplied each sample in a dataset $\mathcal{D}$ by 2.

1 point

- ( ) The variance of the new dataset will be four times the variance of $\mathcal{D}$.
- ( ) The variance of the new dataset will not change.
- ( ) The variance of the new dataset will be two times the variance of $\mathcal{D}$.

5. Assuming we have mean $\bar{x}_{n-1}$ and variance σ^2_{n-1} for some dataset $\mathcal{D}_{n-1}$ with $n-1$ samples. What would be the variance σ^2_n if we add a new element x_* to the dataset (assuming you have computed the new sample mean $\bar{x}_n$) ?

1 point

- ( ) $\sigma^2_n = \frac{n-1}{n}\sigma^2_{n-1} + \frac{1}{n}(x_* - \bar{x}_{n-1})^2$
- ( ) $\sigma^2_n = \frac{n-2}{n-1}\sigma^2_{n-1} + \frac{1}{n}(x_* - \bar{x}_{n-1})(x_* - \bar{x}_n)$
- ( ) $\sigma^2_n = \frac{n-1}{n}\sigma^2_{n-1} + \frac{1}{n}(x_* - \bar{x}_{n-1})(x_* - \bar{x}_n)$
- ( ) $\sigma^2_n = \frac{n-1}{n}\sigma^2_{n-1} + \frac{1}{n-1}(x_* - \bar{x}_{n-1})(x_* - \bar{x}_n)$

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